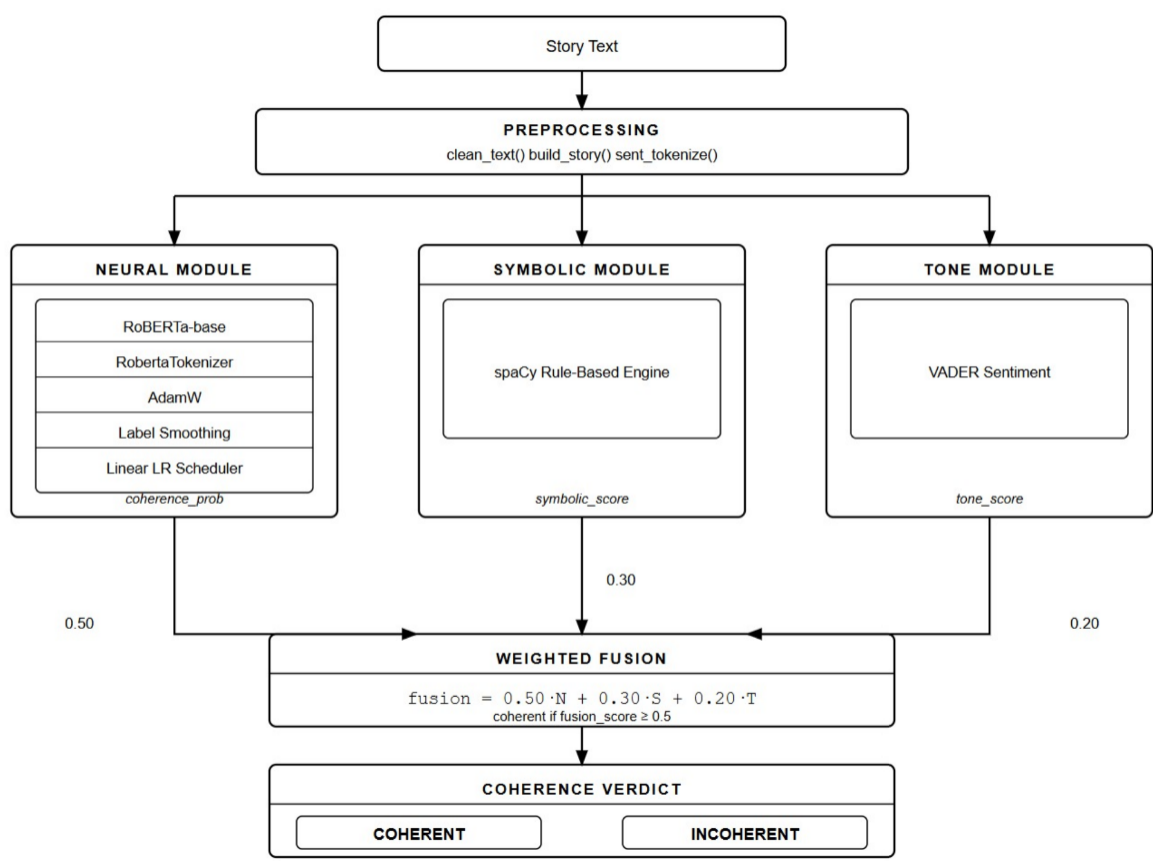


Architecture Document

ChronoMind: A Neuro-Symbolic Engine for Timeline and Logic Verification in Stories and novels

1. System Architecture Diagram

Diagram:



Explanation:

The system architecture follows a modular neuro-symbolic architecture, breaking the system into multiple specialized components:

- **Text Ingestion Module:**

Handles user-uploaded long-form text such as novels and narratives.

- **Preprocessing Engine:**

Performs sentence segmentation, tokenization, and entity extraction to structure raw text.

- **LLM Engine :**

Provides contextual and semantic understanding of the narrative.

- **Symbolic Reasoning Engine:**

Applies rule-based logic for interpretable and explainable reasoning.

- **Temporal Analysis Service:**

Detects inconsistencies and contradictions in event timelines.

- **Logical Consistency Service:**

Identifies logical contradictions within the narrative.

- **Literary Tone Analysis Service:**

Analyzes tone, style, and literary characteristics of the text.

- **Results Aggregator:**

Combines outputs from all analysis services into structured results.

- **User Interface / Dashboard:**

Displays analysis results, metrics, and reports to the user.

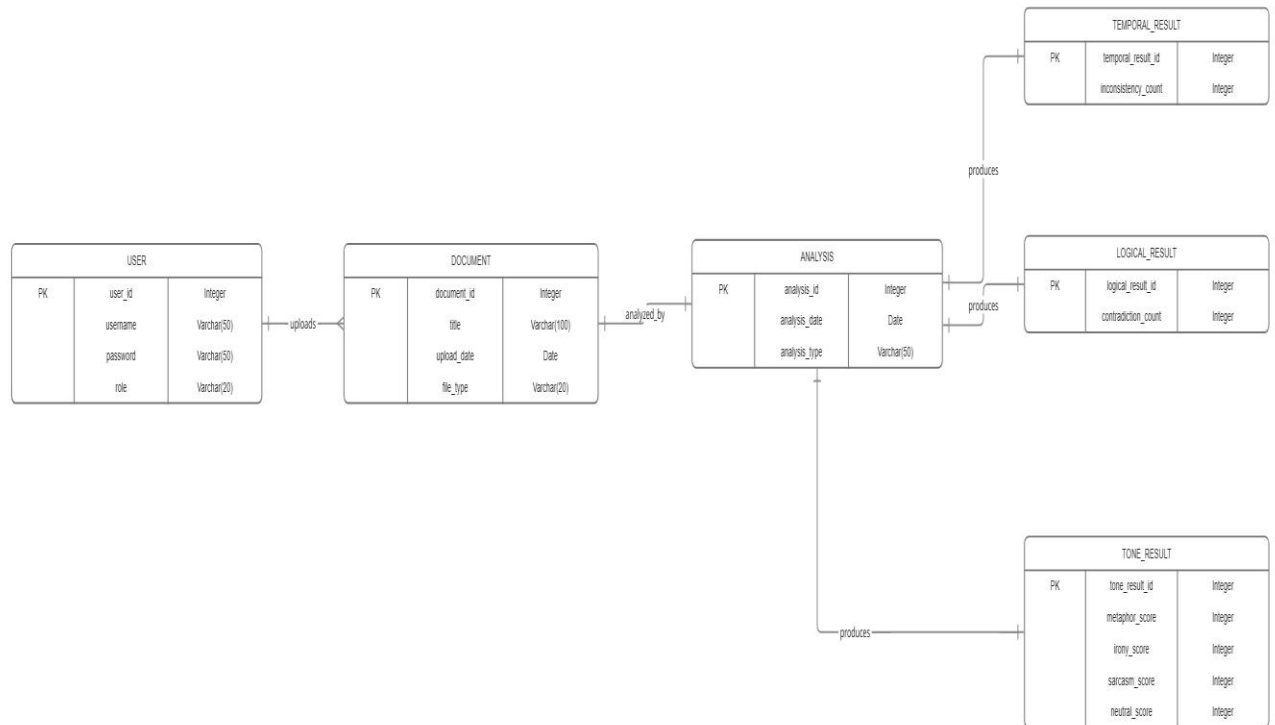
Databases:

- **Text Storage:** Stores processed textual data
- **Analysis Results Database:** Stores detected inconsistencies and tone analysis results

This architecture ensures scalability, interpretability, and efficient processing of long-form documents by combining Large Language Models with symbolic reasoning technique.

2. ER Diagram

Diagram:



Explanation:

The Entity-Relationship (ER) Diagram represents the database structure of the **ChronoMind** system.

The key entities in the system are:

User:

Stores user-related information such as user_id, username, password, and role. Users are responsible for uploading documents into the system.

Document:

Contains information about uploaded text files, including document_id, title, upload_date, and file_type. Each document is uploaded by a user.

Analysis:

Represents the analysis process applied to a document. It stores analysis_id, analysis_date, and analysis_type. Each document is analyzed through this entity.

Temporal_Result:

Stores the results of temporal inconsistency detection. It includes temporal_result_id and inconsistency_count, indicating the number of temporal ambiguities found during analysis.

Logical_Result:

Stores logical inconsistency detection results. It contains logical_result_id and contradiction_count, representing the number of logical contradictions identified.

Tone_Result:

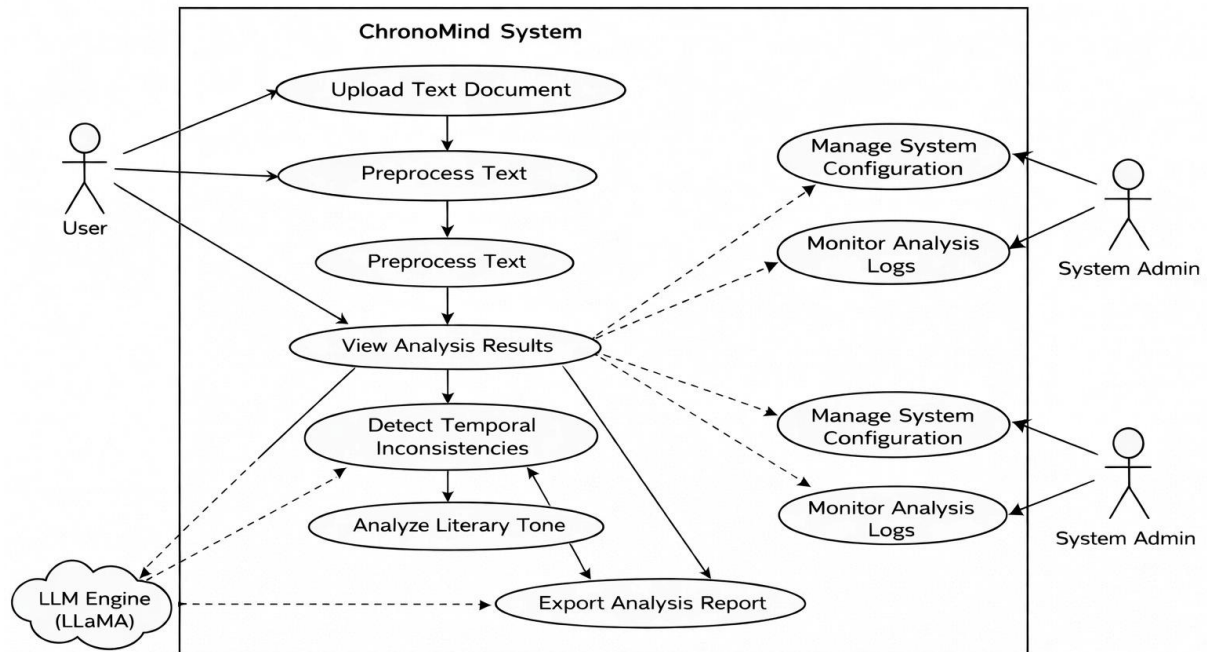
Stores literary tone analysis results, including metaphor_score, irony_score, sarcasm_score, and neutral_score.

Relationships:

- 1) A User uploads Document.
- 2) A Document is analyzed by Analysis.
- 3) An Analysis produces Temporal_Result.
- 4) An Analysis produces Logical_Result.
- 5) An Analysis produces Tone_Result.

3. Use Case Diagram

Diagram:



Explanation:

The Use Case Diagram illustrates the interaction between different actors and the ChronoMind system. The primary actors involved are:

- **User:** Uploads text documents, initiates preprocessing, views analysis results, and exports the final analysis report.
- **LLM Engine :** Assists the system in detecting temporal inconsistencies and analyzing literary tone through advanced language understanding.
- **System Admin:** Manages system configuration and monitors analysis logs to ensure proper system functioning.

Each use case represents a key functionality of ChronoMind, enabling structured text analysis, inconsistency detection, tone evaluation, and report generation in an interpretable and efficient manner.

4. Conclusion

The ChronoMind system integrates neuro-symbolic reasoning with Large Language Models to provide interpretable and scalable analysis of long-form textual narratives. Its modular architecture, well-defined data model, and clear user interactions ensure accurate detection of temporal, logical, and tonal inconsistencies. Overall, the system offers a robust and explainable solution for advanced literary and narrative analysis.